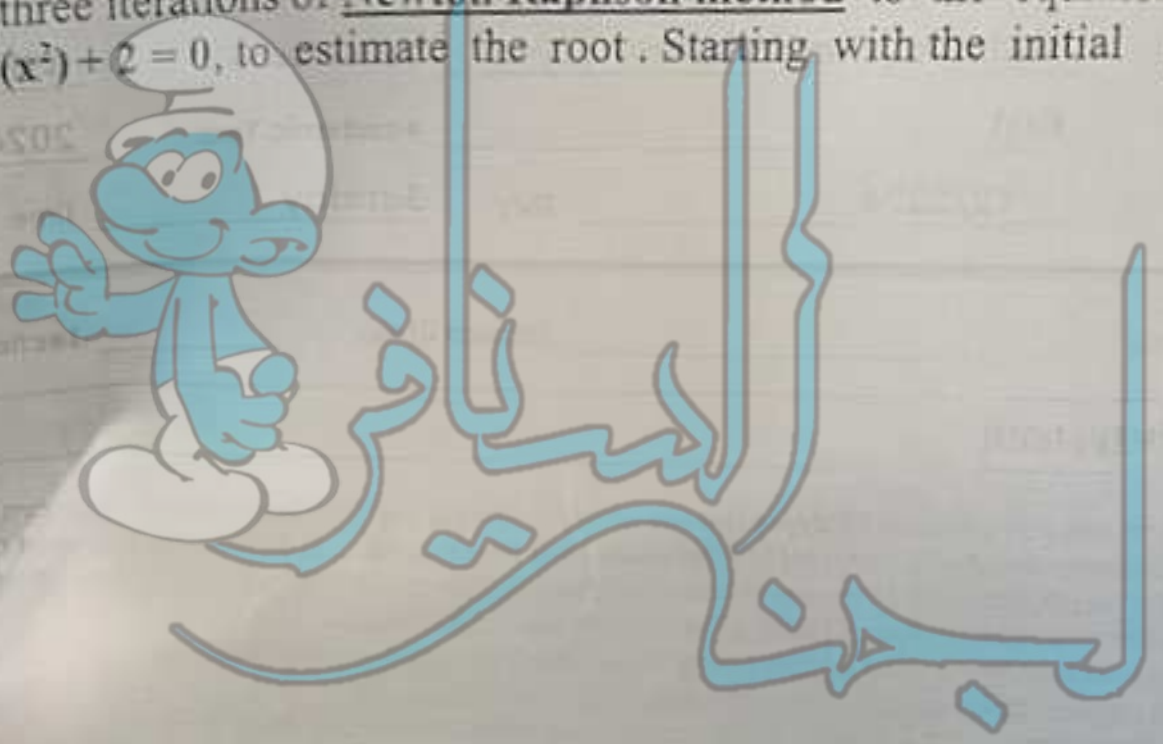
Carry your intermediate computations to at least four decimal places.

Q1- Apply three iterations of **Newton Raphson method** to the equation $-4x + \cos(x^2) + 2 = 0$, to estimate the root . Starting with the initial guess $x_0 = 0.5$.

(10 Marks)



Q2: a) Solve Equations $2x+5y=21$, (4 Marks)

$x+2y=8$ using *Cramer's Rule method*



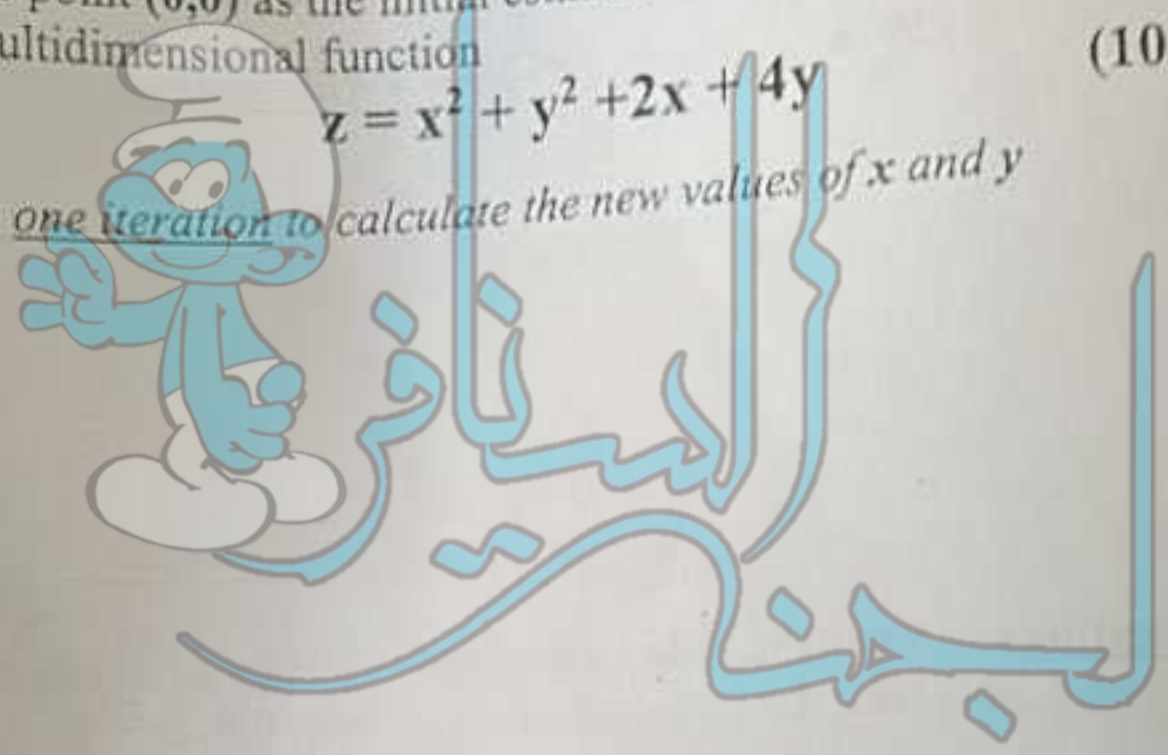
b) Perform **two iterations** to solve the following system of linear equations using the iterative Jacobi method. Make the initial guess $x_1 = 0, x_2 = 0, x_3 = 0$. (6Marks)

$$8x_1 - 2x_2 + 3x_3 = -16$$
$$-3x_1 + 9x_2 + x_3 = 24$$
$$2x_1 - x_2 - 6x_3 = 36$$

Q3- Use the point **(0,0)** as the initial estimate of the *optimal gradient steepest method* to the multidimensional function **(10 Marks)**

$$z = x^2 + y^2 + 2x + 4y$$

Perform one iteration to calculate the new values of x and y



NO Date

محمد ماهر الحلوح.

$Q_{(1)}$:- 3 iteration :- N.R. Method , $f(x) = -4x + \cos(x^2) + 2 = 0$, $x_0 = 0.5$

sol :- $x_{i+1} = x_i - \frac{f(x)}{f'(x)}$ $\quad f'(x) = -4 - 2x \sin(x^2) = 0$

iteration[1] :-

$$x_1 = 0.5 - \frac{-4(0.5) + \cos(0.5^2) + 2}{-4 - 2(0.5)\sin(0.5^2)} = \boxed{0.7281187} \rightarrow x_1$$

iteration[2]

$$x_2 = 7.281187 - \frac{-4(0.7281187) + \cos(0.7281187) + 2}{-4 - 2(0.7281187)\sin(0.7281187^2)}$$

$$= \boxed{0.7176155} \rightarrow x_2$$

iteration[3]

$$x_3 = 0.7176155 - \frac{-4(0.7176155) + \cos(0.7176155) + 2}{-4 - 2(0.7176155)\sin(0.7176155^2)}$$

$$= \boxed{0.7175824} \rightarrow x_3 \qquad x_4 = 0.7175824$$

محمد الحلوح

$Q_{(2)}$:- (0, 0) , optimal gradient steepest method one iteration.

$f(x) = x^2 + y^2 + 2x + 4y$

sol :- $df/dx = 2x + 0 + 2 + 0 = 2x + 2 = 0$ $\quad \nabla f(0,0) = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

$df/dy = 0 + 2y + 0 + 4 = 2y + 4 = 0$

$x_{i+1} = 0 + 2h = 2h$

$y_{i+1} = 0 + 4h = 4h$

$g(x) = f(2h, 4h)$

$g(x) = 4h^2 + 16h^2 + 4h + 16h$

$g(x) = 20h^2 + 20h$

$g'(x) = 40h + 20 = 0 \Rightarrow \frac{40h}{40} = \frac{-20}{40}$

$\therefore \boxed{h = -0.5}$

$\therefore x_2 = 2h = 2 * -0.5$

$= -1$

$y_2 = 4h = 4 * -0.5$

$= -2$ $\quad \therefore (-1, -2)$

$h = 2$ $\quad f(x) = 2x^2 + 2x - 2$ $\quad f'(x) = 4x + 2$

True value = 22 $\quad f(x_{i+1}) = f(1) + 8$

$2 + 6 + 2$

$2 + 12 = 14$

$\varepsilon_0 = 22 - 14 = \boxed{8}$

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Q(3): $2x + 5y = 21$

$x + 2y = 8$ using Cramer's Rule method.

Sol

$$[A]\begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 21 \\ 8 \end{bmatrix}$$

$$x_1 = \frac{\begin{bmatrix} 21 & 5 \\ 8 & 2 \end{bmatrix}}{\det A} = \frac{\begin{bmatrix} 21 & 5 \\ 8 & 2 \end{bmatrix}}{-1} = \frac{(21*2)-(5*8)}{-1}$$

$$= -2$$

$$\det[A] = (2*2)-(5*1) = -1$$

$$x_2 = \frac{\begin{bmatrix} 2 & 21 \\ 1 & 8 \end{bmatrix}}{-1} = \frac{(2*8)-(21*1)}{-1} = 5$$

$x = -2$ / $y = 5$

Q(4): two iterations: Jacobi method, $x_1 = 0$, $x_2 = 0$, $x_3 = 0$

$8x_1 - 2x_2 + 3x_3 = -16$

$-3x_1 + 9x_2 + x_3 = 24$

$2x_1 - x_2 - 6x_3 = 36$

Sol

$$x_1 = f(x_2, x_3) = \frac{1}{a_{11}}(b_1 - a_{12}x_1 - a_{13}x_2)$$

$$x_1 = \frac{1}{8}(-16 - 0 - 0) = \boxed{-2}$$

$$x_2 = f(x_1, x_3) = \frac{1}{a_{22}}(b_2 - a_{21}x_1 - a_{23}x_3)$$

$$x_2 = \frac{1}{9}(24 - 0 - 0) = \boxed{\frac{8}{3}}$$

$$x_3 = f(x_1, x_2) = \frac{1}{a_{33}}(b_3 - a_{31}x_1 - a_{32}x_2)$$

$$x_3 = \frac{1}{-6}(36 - 0 - 0) = \boxed{-6}$$

$$\begin{bmatrix} 8 & -2 & 3 \\ -3 & 9 & 1 \\ 2 & -1 & -6 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leftarrow \text{iteration (1)}$$

$$= \begin{bmatrix} -16 \\ 24 \\ 36 \end{bmatrix}$$

iteration [2] →

$x_1 = -2$, $x_2 = \frac{8}{3}$, $x_3 = -6$

$$x_1 = \frac{1}{8}\left(-16 - (-2*-2) - (3*\tfrac{8}{3})\right) = -3.5$$

$$x_2 = \frac{1}{9}\left(24 - (-3*-2) - (1*-6)\right) = \frac{8}{3} = 2.6667$$

$$x_3 = \frac{1}{-6}\left(36 - (2*-2) - (-1*\tfrac{8}{3})\right) = -7.111$$

$f(1) = 5$, $f'(1) = 3$, $f''(1) = 2$, $f'''(1) = -3$ (x_i)

$h = 1$

$h = x_{i+1} - x_i$

$x_{i+1} = 1 + 1 = \boxed{2}$

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \frac{f'''(x_i)h^3}{3!} + \frac{f^{IV}(x_i)h^4}{4!}$$

$$= 5 + 6 + 4 + -4$$

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